

- Q.30 Draw the cross section of a hill road partially in cutting and partially in filling.
- Q.31 What are the main causes responsible for failures in wearing course of a flexible pavement?
- Q.32 Explain superelevation.
- Q.33 Explain runways. What are the factors affecting direction of runway?
- Q.34 Explain continuous bay method of road construction
- Q.35 Define prime coat, seal coat and tack coat.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Name the various equipment used for construction. Explain any one with neat sketch.
- Q.37 What are the various method of soil stabilization? Explain mechanical stabilization in detail.
- Q.38 Explain CBR test to determine the strength of soil subgrade.

No. of Printed Pages : 4 180752/170752/120752
Roll No. /030752

5th Sem / Civil., Constr. Mgmt., Civil Engg (Spl Highway Engg)

Subject:- Highway Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Milestone painted in green top and white base depict
a) National Highway b) State Highway
c) District Roads d) Expressway
- Q.2 Design speed for National Highway is
a) 80 km/hr b) 70 km / hr
c) 60 km / hr d) 50 km / hr
- Q.3 A horizontal curve whose radius changes gradually from an infinity to same finite value
a) Simple Curve b) Compound Curve
c) Transition Curve d) Reverse Curve
- Q.4 After reconnaissance, the road survey is conducted is termed as
a) Map study b) Location survey
c) Preliminary Survey d) Detailed Survey
- Q.5 The strength of aggregate is assessed by
a) Impact Test b) Crushing Value Test
c) Abrasion Test d) Soundness Test

- Q.6 Application of low viscosity bitumen over porous untreated surface is termed as
 a) Prime Coat b) Base coat
 c) Wearing Coat d) Subsurface Course
- Q.7 The drains provided below the road bed for cross drainage of water are called
 a) Catch Water Drain b) Side Drain
 c) Cross Drain d) Under Drain
- Q.8 In hot mix plant, mixing of bitumen and aggregates is carried in a
 a) Pug Mixer b) Tar Boiler
 c) Asphalt Mixer d) Grader
- Q.9 The best site for an airport is
 a) hill b) Valley
 c) Hill Top d) Mountain
- Q.10 Rollers used to compact clayey soil
 a) Sheep Foot Roller
 b) Pneumatic Type Roller
 c) Flat Wheeled Roller
 d) Tandem Roller

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 NHAI means_____.
- Q.12 The minimum shoulder width recommended by IRC is _____.
- Q.13 The road alignment between two stations should be as _____ as possible.

(2) 180752/170752/120752
/030752

- Q.14 Softening point test is conducted by_____ apparatus
- Q.15 Alternate bay method is used for the construction of_____ roads.
- Q.16 The curve in a hill road which changes its direction through an angle of 180° or so is known as_____.
- Q.17 The water entering the road bed materials decreases their_____.
- Q.18 The deformation of flexible pavement due to consolidation of underlying layers refers to the formation of_____.
- Q.19 _____ is used for digging trench.
- Q.20 AAI stands for_____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define CRRI. State its functions.
- Q.22 Define gradient. What is the purpose of providing gradient.
- Q.23 Explain stopping sight distance and overtaking sight distance.
- Q.24 Define road Shoulders, kerbs and footpath.
- Q.25 What are the requirement of an ideal road alignment?
- Q.26 What are the various tests to be performed on road aggregates.
- Q.27 Comparison between flexible and rigid pavement.
- Q.28 What are the requirement of a good road aggregate?
- Q.29 Explain borrow pits and spoil banks.

(3) 180752/170752/120752
/030752